

$$4) \quad 4.5) \quad a - \quad g(\{1, 2, 3, 4\}) = \left\{1, \frac{3}{2}, \frac{5}{3}, \frac{7}{4}\right\};$$

$$g^{-1}(\{1, 5\}) = \{1\}$$

$$b - \quad f(\{x \in \mathbb{R} \mid x^2 - 16 = 0\})$$

$$= f^{-1}(\{10\}) = \{x \in \mathbb{R} : f(x) = 10\}$$

$$= [4, 20] \cup]30, +\infty[$$